

US Patent & Trademark Office

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United States Patent Application Publication

20250285728

Kind Code

A1

Publication Date

September 11, 2025

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PRESCRIPTION ASSISTANCE DEVICE AND PRESCRIPTION ASSISTANCE METHOD

Abstract

A prescription assistance device for assisting determination an appropriate prescription of drugs for treatment of heart failure from a variety of combinations of a type, an administration amount, and an administration timing of the drugs, which includes a processor configured to: acquire attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generate, receive, or read, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and predict an effect of the drug for each of a plurality of prescription candidates.

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Appl. No.: 19/214265

Filed: May 21, 2025

Foreign Application Priority Data

JP	2020-054748	Mar. 25, 2020
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Related U.S. Application Data

parent US continuation 17889653 20220817 parent-grant-document US 12334201 child US 19214265

Publication Classification

Int. Cl.: G16H20/10 (20180101); G16H10/60 (20180101)

U.S. Cl.:

CPC G16H20/10 (20180101); G16H10/60 (20180101);

Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/889,653 filed on Aug. 17, 2022, which is a continuation of International Application No. PCT/JP2021/009084 filed on Mar. 8, 2021, which claims priority to Japanese Application No. 2020-054748 filed on Mar. 25, 2020, the entire content of all three of which is incorporated herein by reference.

TECHNOLOGICAL FIELD

[0002] The present disclosure generally relates to a prescription assistance device and a prescription assistance method.

BACKGROUND DISCUSSION

[0003] Japanese Patent Application Publication No. 2019-095960 A describes a treatment selection assistance system that predicts, for each treatment unit, an achievement level of a treatment target for a patient, calculates an appropriateness of each treatment unit for the patient, and provides information on a treatment unit suitable for the patient based on the predicted achievement level and the calculated appropriateness.

[0004] Treatment on acute heart failure can be performed with a drug such as a diuretic, a cardiotonic, and an infusion. In order to release systemic congestion, a plurality of types of diuretics are combined in accordance with a cardiac function and a renal function, a cardiotonic is used to ensure a urine volume and to help prevent dehydration, and the diuretics, the cardiotonic, and the infusion are adjusted so as to maintain an electrolyte balance, albumin, and the like.

[0005] Pathology of patients with heart failure is very diverse. Although a prescription of drugs is determined by an experience of a doctor, it can be difficult to determine the prescription after completely grasping a state of a patient. In addition, when there are a variety of combinations of a type, an administration amount, and an administration timing of the drug as in the treatment of heart failure, it is necessary to determine a complicated prescription. In a system in the related art, it is difficult to assist determination on such a complicated prescription of drugs.

SUMMARY

[0006] The present disclosure is to assist determination on a relatively complicated prescription of drugs.

[0007] A prescription assistance device as an aspect of the present disclosure includes a processor configured to acquire attribute data indicating an attribute of a patient, time series data on a state of the patient, and drug history data indicating a prescription history of a drug for the patient, configured to predict, based on the attribute data, the time series data, and the drug history data, an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another, and configured to present a prediction result to a user of the effect of the drug for each of the plurality of prescription candidates.

[0008] In an embodiment, the time series data includes results of vital measurement on the patient at a plurality of time points, and the processor is configured to predict, as the effect of the drug, results of vital measurement on the patient after drug administration.

[0009] In an embodiment, the vital measurement on the patient includes urine volume measurement on the patient.

[0010] In an embodiment, the time series data includes results of blood tests on the patient at a plurality of time points, and the processor is configured to predict, as the effect of the drug, results of blood tests on the patient after drug administration.

[0011] In an embodiment, the blood test on the patient includes at least one of a cardiac function test, a renal function test, and an electrolyte test on the patient.

[0012] In an embodiment, the processor is configured to predict, as the effect of the drug, the number of days until the patient leaves a bed after drug administration.

[0013] In an embodiment, the drug includes at least one of a diuretic, a vasodilator, a cardiotonic, a vasopressor, a heart rate controlling drug, an analgesic, and an infusion.

[0014] In an embodiment, the processor is configured to receive, from the user, an operation for adjusting at least one of the plurality of prescription candidates.

[0015] In an embodiment, the processor is configured to receive, from the user, an operation for selecting any one of the plurality of prescription candidates, and controls, according to the selected prescription candidate, a device configured to administer the drug to the patient.

[0016] In an embodiment, the processor is configured to predict the effect of the drug using a trained model to which the attribute data, the time series data, and the drug history data are input and from which the effect of the drug is output.

[0017] In an embodiment, when the drug is administered to the patient, the processor is configured to acquire data on the state of the patient at a timing determined for each type of the drug, and to feed the data back to the trained model.

[0018] In an embodiment, when a therapeutic intervention different from the drug administration is performed on the patient, the processor is configured to exclude at least a part of the time series data before the therapeutic intervention and to predict the effect of the drug.

[0019] In an embodiment, the therapeutic intervention includes at least one of respiratory support and blood circulation support.

[0020] A prescription assistance method as an aspect of the present disclosure includes: acquiring, by a prescription assistance device, attribute data indicating an attribute of a patient, time series data on a state of the patient, and drug history data indicating a prescription history of a drug for the patient; predicting, by the prescription assistance device, an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another based on the attribute data, the time series data, and the drug history data; and presenting, by the prescription assistance device, a prediction result to a user of the effect of the drug for each of the plurality of prescription candidates.

[0021] A prescription assistance method as another aspect of the present disclosure includes: acquiring attribute data indicating an attribute of a patient, time series data on a state of the patient, and drug history data indicating a prescription history of a drug for the patient; predicting an effect of the drug for a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another based on the attribute data, the time series data, and the drug history data; and displaying a prediction result for the effect of the drug for the plurality of prescription candidates in which combinations of the type, the administration amount, and the administration timing of the drug are different from one another based on the attribute data, the time series data, and the drug history data.

[0022] A prescription assistance device for assisting determination an appropriate prescription of

drugs for treatment of heart failure from a variety of combinations of a type, an administration amount, and an administration timing of the drugs comprising: a processor configured to: acquire attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generate, receive, or read, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; predict at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure, based on the attribute data, the time series data, and the drug history data, as an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and present a prediction result of at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure to a user for each of the plurality of prescription candidates via an output interface for determination by the user an appropriate prescription of drugs as in the treatment of heart failure.

[0023] A prescription assistance method for assisting determination an appropriate prescription of drugs for treatment of heart failure from a variety of combinations of a type, an administration amount, and an administration timing of the drugs comprising: acquiring attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generating, receiving, or reading, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; predicting at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure, based on the attribute data, the time series data, and the drug history data, as an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and presenting a prediction result of at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure to a user for each of the plurality of prescription candidates via an output interface for determination by the user an appropriate prescription of drugs as in the treatment of heart failure.

[0024] A non-transitory computer-readable medium storing a computer program code executed by a computer processor that executes a process for a prescription assistance device for assisting determination an appropriate prescription of drugs for treatment of heart failure from a variety of combinations of a type, an administration amount, and an administration timing of the drugs, the process comprising: acquiring attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generating, receiving, or reading, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; predicting at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure, based on the attribute data, the time series data, and the drug history

data, as an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and presenting a prediction result of at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure to a user for each of the plurality of prescription candidates via an output interface for determination by the user an appropriate prescription of drugs as in the treatment of heart failure.

[0025] According to the present disclosure, it is possible to assist determination on a complicated prescription of drugs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a diagram showing a configuration of a system as an aspect of the present disclosure.

[0027] FIG. 2 is a diagram showing functions of the system as an aspect of the present disclosure and a flow of data in the system.

[0028] FIG. 3 is a block diagram showing details of a prescription candidate generation function of a prescription assistance device according to a first embodiment.

[0029] FIG. 4 is a flowchart showing an operation of the prescription assistance device according to the first embodiment.

[0030] FIG. 5 is a block diagram showing details of a prescription candidate generation function of a prescription assistance device according to a second embodiment.

[0031] FIG. 6 is a flowchart showing an operation of the prescription assistance device according to the second embodiment.

DETAILED DESCRIPTION

[0032] Set forth below with reference to the accompanying drawings is a detailed description of embodiments of a prescription assistance device and a prescription assistance method. Note that since embodiments described below are preferred specific examples of the present disclosure, although various technically preferable limitations are given, the scope of the present disclosure is not limited to the embodiments unless otherwise specified in the following descriptions.

[0033] In the drawings, the same or corresponding parts are denoted by the same reference numerals. In the description of the embodiments, the description of the same or corresponding parts will be omitted or simplified as appropriate.

[0034] A configuration of a system **10** as an aspect of the present disclosure will be described with reference to FIG. 1.

[0035] The system **10** shown in FIG. 1 includes a prescription assistance device **20**, a database **30**, a pump **40**, and a sensor **50**.

[0036] The prescription assistance device **20** is communicably connected to the database **30**, the pump **40**, and the sensor **50** directly or via a network such as a local area network (LAN).

[0037] The prescription assistance device **20** can be installed in a hospital as an example, and may be installed in another facility such as a data center. The prescription assistance device **20** can be, for example, a general-purpose computer such as a personal computer (PC) or a server computer, or a dedicated computer.

[0038] The database **30** can be installed in a hospital as an example, and may be installed in another facility such as a data center. The database **30** can be, for example, a relational database management system (RDBMS). The database **30** can be separate from the prescription assistance device **20** as an example, or alternatively, may be integrated with the prescription assistance device **20**.

[0039] The pump **40** may be installed in a hospital. The pump **40** can be, for example, an infusion pump or a syringe pump. The pump **40** may be a smart pump.

[0040] The sensor **50** may be installed in a hospital. The sensor **50** can be, for example, a percutaneous oxygen saturation (SpO₂) sensor, a heart rate sensor, a blood pressure sensor, a weight scale, a urine volume sensor, a body temperature sensor, a pulse sensor, or a respiration sensor.

[0041] A configuration of the prescription assistance device **20** will be described with reference to FIG. **1**.

[0042] The prescription assistance device **20** can include a control unit **21**, a storage unit **22**, a communication unit **23**, an input unit **24**, and an output unit **25**.

[0043] The control unit **21** can include at least one processor, at least one dedicated circuit, or a combination of at least one processor and at least one dedicated circuit. The processor may be a general-purpose processor such as a central processing unit (CPU) or a graphics processing unit (GPU), or a dedicated processor specialized for specific processing. The dedicated circuit can be, for example, a field-programmable gate array (FPGA) or an application specific integrated circuit (ASIC). The control unit **21** executes processing related to an operation of the prescription assistance device **20** while controlling the units of the prescription assistance device **20**.

[0044] The storage unit **22** includes at least one semiconductor memory, at least one magnetic memory, at least one optical memory, or a combination of at least two types of the memories. The semiconductor memory can be, for example, a random access memory (RAM) or a read only memory (ROM). The RAM can be, for example, a static random access memory (SRAM) or a dynamic random access memory (DRAM). The ROM can be, for example, an electrically erasable programmable read only memory (EEPROM). The storage unit **22** functions as, for example, a main storage device, an auxiliary storage device, or a cache memory. The storage unit **22** stores data used for the operation of the prescription assistance device **20** and data obtained by the operation of the prescription assistance device **20**. The database **30** may be integrated with the prescription assistance device **20** by being constructed in the storage unit **22**.

[0045] The communication unit **23** includes at least one communication interface. The communication interface can be, for example, a LAN interface. The communication unit **23** receives the data used for the operation of the prescription assistance device **20** and transmits the data obtained by the operation of the prescription assistance device **20**.

[0046] The input unit **24** includes at least one input interface. The input interface can be, for example, a physical key, a capacitive key, a pointing device, a touch screen provided integrally with a display, or a microphone. The input unit **24** receives an operation for inputting the data used for the operation of the prescription assistance device **20**. The input unit **24** may be connected to the prescription assistance device **20** as an external input device instead of being provided in the prescription assistance device **20**. For example, any method such as a universal serial bus (USB), a high-definition multimedia interface (HDMI®), or Bluetooth® can be used as a connection method.

[0047] The output unit **25** includes at least one output interface. The output interface can be, for example, a display or a speaker. The display can be, for example, a liquid crystal display (LCD) or an organic electro luminescence (EL) display. The output unit **25** outputs the data obtained by the operation of the prescription assistance device **20**. The output unit **25** may be connected to the prescription assistance device **20** as an external output device instead of being provided in the prescription assistance device **20**. For example, any method such as USB, HDMI, or Bluetooth can be used as a connection method.

[0048] A function of the prescription assistance device **20** is implemented by executing a program as an aspect of the present disclosure by a processor corresponding to the control unit **21**. That is, the function of the prescription assistance device **20** can be implemented by software. The program causes a computer to function as the prescription assistance device **20** by causing the computer to

execute the operation of the prescription assistance device **20**. That is, the computer functions as the prescription assistance device **20** by executing the operation of the prescription assistance device **20** according to the program.

[0049] The program can be stored in a non-transitory computer-readable medium. The non-transitory computer-readable medium can be, for example, a flash memory, a magnetic recording device, an optical disc, a magneto-optical recording medium, or a ROM. Distribution of the program is executed by, for example, selling, transferring, or lending a portable medium such as secure digital (SD) card, a digital versatile disc (DVD), or a compact disc read only memory (CD-ROM) in which the program is stored. The program may be distributed by storing the program in a storage of a server and transferring the program from the server to another computer. The program may be provided as a program product.

[0050] For example, the computer temporarily stores, in a main storage device, the program stored in the portable medium or the program transferred from the server. Then, the computer reads the program stored in the main storage device by the processor, and executes processing according to the read program by the processor. The computer may read the program directly from the portable medium and execute processing according to the program. Each time the program is transferred from the server to the computer, the computer may sequentially execute processing according to the received program. The processing may be executed by a so-called application service provider (ASP) type service in which the function is implemented only by instruction execution and result acquisition without transferring the program from the server to the computer. The program includes information that is provided for processing by an electronic computer and that is treated equivalent to a program. For example, data that is not a direct command to the computer but has a property that defines processing of the computer corresponds to “information that is treated equivalent to a program”.

[0051] A part or all of the functions of the prescription assistance device **20** may be implemented by a dedicated circuit corresponding to the control unit **21**. That is, a part or all of the functions of the prescription assistance device **20** may be implemented by hardware.

[0052] A function of the system **10** as an aspect of the present disclosure and a flow of data in the system **10** will be described with reference to FIG. 2.

[0053] The prescription assistance device **20** has a prescription candidate generation function **61** and an effect prediction function **62**.

[0054] Attribute data **71** indicating an attribute of a patient, time series data **72** on a state of the patient, and drug history data **73** indicating a prescription history of a drug for the patient are registered with the database **30**.

[0055] The attribute data **71** may be input from another system or manually input. The attribute data **71** can include, for example, data indicating at least one of age, sex, blood type, and comorbidity, as the attribute of the patient.

[0056] The time series data **72** includes sensor data **74**, test data **75**, and manually input data **76**. The sensor data **74** can be input from the sensor **50**. The sensor data **74** can include, for example, data indicating at least one of SpO₂, a heart rate, a cardiac output, a cardiac index, a blood pressure, a body weight, a left ventricular end diastolic pressure, a left ventricular ejection fraction, a central venous pressure, a urine volume, a respiratory rate, a pulmonary artery wedge input pressure, and a body water content, as the state of the patient. The test data **75** may be input, for example, from another system or manually input. The test data **75** can include, for example, data indicating at least one of a liver function, a renal function, creatinine, estimated glomerular filtration rate (eGFR), brain natriuretic peptide (BNP), N-terminal fragment of pro-brain natriuretic peptide (NT-proBNP), hemoglobin, sodium, potassium, blood urea nitrogen (BUN), albumin, and C-reactive protein (CRP), as the state of the patient. The manually input data **76** is manually input. The manually input data **76** can include, for example, data indicating at least one of presence or absence of edema such as leg edema, presence or absence of cold extremities, presence or absence

of cyanosis, presence or absence of jugular venous distension, presence or absence of orthopnea, the urine volume, and the body weight, as the state of the patient.

[0057] The drug history data **73** can be input from another system or manually input. The drug history data **73** can include data indicating at least one of a type, an administration amount, an administration method, an administration start date and time, and an administration end date and time of the drug.

[0058] An electronic medical record **77** is further registered with the database **30**. At least one of the attribute data **71**, the time series data **72**, and the drug history data **73** may be registered as a part of the electronic medical record **77**.

[0059] The prescription assistance device **20** acquires, from the database **30**, the attribute data **71**, the time series data **72**, the drug history data **73**, and the electronic medical record **77**. The prescription assistance device **20** may directly acquire, from the sensor **50**, the sensor data **74** provided in the time series data **72**.

[0060] The control unit **21** of the prescription assistance device **20** generates, by the prescription candidate generation function **61**, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another. The control unit **21** predicts, by the effect prediction function **62**, the effect of the drug for each of the plurality of generated prescription candidates, based on the attribute data **71**, the time series data **72**, and the drug history data **73**. The control unit **21** presents a calculation result **78** which is a prediction result to a user such as a doctor.

[0061] According to the above aspects, it is possible to assist determination on a relatively complicated prescription. That is, even when there are a variety of combinations of the type, the administration amount, and the administration timing of the drug are manifold as in treatment on heart failure, it is possible to notify the user of the effect when each combination is applied to the patient and to rather easily determine the prescription.

[0062] The prescription candidate generation function **61** may be omitted. For example, instead of generating the plurality of prescription candidates, the control unit **21** of the prescription assistance device **20** may receive an input of a plurality of prescription candidates from the user via the input unit **24** such as a touch screen. Alternatively, the control unit **21** may receive, via the communication unit **23**, a plurality of prescription candidates devised by other doctors, which are shared by an external system such as a cloud computing system. Alternatively, the control unit **21** may read a plurality of prescription candidates stored in advance in the storage unit **22**.

[0063] Hereinafter, several embodiments as specific examples of the aspects shown in FIGS. **1** and **2** will be described with reference to the drawings.

First Embodiment

[0064] In the present embodiment, as shown in FIG. **3**, the prescription candidate generation function **61** of the prescription assistance device **20** can include a prescription range setting function **63**, a prescription range storage function **64**, a previous prescription input function **65**, a prescription candidate creation function **66**, and a prescription candidate output function **67**.

[0065] An operation of the prescription assistance device **20** according to the present embodiment will be described with reference to FIG. **4**. The operation corresponds to a prescription assistance method according to the present embodiment.

[0066] In **S101**, the control unit **21** acquires the attribute data **71**.

[0067] In the present embodiment, the control unit **21** receives an input of the attribute data **71** via the database **30**. Specifically, the control unit **21** receives, from the database **30** via the communication unit **23**, the attribute data **71** input to the database **30**.

[0068] As a modification of the present embodiment, the control unit **21** may receive an input of the attribute data **71** from a user via the input unit **24** such as a touch screen.

[0069] In **S102**, the control unit **21** acquires the time series data **72**.

[0070] In the present embodiment, the control unit **21** receives an input of the time series data **72**

via the database **30**. Specifically, the control unit **21** receives, from the database **30** via the communication unit **23**, the sensor data **74**, the test data **75**, and the manually input data **76** that are input to the database **30**.

[0071] As a modification of the present embodiment, the control unit **21** may directly receive an input of the sensor data **74** from the sensor **50**. Specifically, the control unit **21** may directly receive, from the sensor **50** via the communication unit **23**, the sensor data **74** obtained by the sensor **50**. Alternatively, the control unit **21** may receive, from the user via the input unit **24** such as a touch screen, an input of the manually input data **76** and other data that are provided in the time series data **72**.

[0072] In **S103**, the control unit **21** acquires the drug history data **73**.

[0073] In the present embodiment, the control unit **21** receives an input of the drug history data **73** via the database **30**. Specifically, the control unit **21** receives, from the database **30** via the communication unit **23**, the drug history data **73** input to the database **30**.

[0074] As a modification of the present embodiment, the control unit **21** may receive an input of the drug history data **73** from the user via the input unit **24** such as a touch screen.

[0075] In **S104**, the control unit **21** generates a plurality of prescription candidates by the prescription candidate generation function **61**.

[0076] Specifically, the control unit **21** can set, as a prescription range by the prescription range setting function **63**, a restriction range of a type, an administration amount, and an administration timing of the drug that may be prescribed. The control unit **21** stores the set prescription range in the storage unit **22** by the prescription range storage function **64**. The control unit **21** receives, from the user via the input unit **24** such as a touch screen by the previous prescription input function **65**, an input of a prescription previously applied to a patient. The control unit **21** creates, based on the input previous prescription by the prescription candidate creation function **66**, a plurality of prescription candidates by combining the type, the administration amount, and the administration timing of the drug in a plurality of patterns such that the combinations fall within the prescription range stored in the storage unit **22**. The control unit **21** outputs the plurality of created prescription candidates by the prescription candidate output function **67**.

[0077] In the present embodiment, when creating a plurality of prescription candidates, the control unit **21** refers to the attribute data **71** acquired in **S101** and adjusts the plurality of prescription candidates to be created according to the attribute of the patient. Alternatively, the control unit **21** may create a plurality of prescription candidates without referring to the attribute data **71**.

[0078] In the present embodiment, when creating a plurality of prescription candidates, the control unit **21** refers to the time series data **72** acquired in **S102** and adjusts the plurality of prescription candidates to be created according to the state of the patient. Alternatively, the control unit **21** may create a plurality of prescription candidates without referring to the time series data **72**.

[0079] In the present embodiment, when creating a plurality of prescription candidates, the control unit **21** refers to the drug history data **73** acquired in **S103** and adjusts the plurality of prescription candidates to be created according to the prescription history of the drug for the patient.

Alternatively, the control unit **21** may create a plurality of prescription candidates without referring to the drug history data **73**.

[0080] As an example, the drug to be prescribed can be a drug used for treatment of heart failure, and specifically can include at least one of a diuretic, a vasodilator, a cardiotonic, a vasopressor, a heart rate controlling drug, an analgesic, and an infusion.

[0081] In **S105**, the control unit **21** predicts, by the effect prediction function **62**, an effect of the drug for each of the plurality of prescription candidates generated in **S104** based on the attribute data **71** acquired in **S101**, the time series data **72** acquired in **S102**, and the drug history data **73** acquired in **S103**.

[0082] Specifically, the control unit **21** can help predict the effect of the drug using a trained model to which the attribute data **71**, the time series data **72**, and the drug history data **73** are input and

from which the effect of the drug is output. The trained model may be constructed by any method, and is constructed by machine learning in the present embodiment. Any method such as recurrent neural network (RNN), support-vector machine (SVM), or random forest can be used as a method of machine learning.

[0083] In the present embodiment, the time series data **72** includes results of vital measurement on the patient at a plurality of time points. The control unit **21** predicts, as the effect of the drug, results of vital measurement on the patient after drug administration. As an example, the vital measurement on the patient includes urine volume measurement on the patient.

[0084] As an example, the results of the vital measurement on the patient can include a numerical value related to heart failure diagnosis. For example, the results of the vital measurement on the patient can include at least one of SpO₂, a heart rate, a cardiac output, a cardiac index, a blood pressure, a body weight, a left ventricular end diastolic pressure, a left ventricular ejection fraction, a central venous pressure, a urine volume, a respiratory rate, a pulmonary artery wedge input pressure, a body water content, a liver function, a renal function, creatinine, eGFR, BNP, NT-proBNP, hemoglobin, sodium, potassium, BUN, albumin, CRP, presence or absence of edema such as leg edema, presence or absence of cold extremities, presence or absence of cyanosis, presence or absence of jugular venous distension, and presence or absence of orthopnea.

[0085] In the present embodiment, the time series data **72** can include results of a blood test on the patient at a plurality of time points. The control unit **21** can predict, as the effect of the drug, results of the blood test on the patient after the drug administration. As an example, the blood test on the patient can include at least one of a cardiac function test, a renal function test, and an electrolyte test on the patient.

[0086] The control unit **21** may predict, as the effect of the drug, the number of days until the patient leaves a bed after the drug administration.

[0087] In **S106**, the control unit **21** can present, to the user, the calculation result **78** which is a prediction result in **S105**.

[0088] Specifically, the control unit **21** can output the calculation result **78** to the output unit **25**, for example, such as a display.

[0089] As an example, it is assumed that the influence of drug administration on the renal function of an acute heart failure patient is predicted. In this case, in **S102**, the sensor data **74** including vital measurement values such as the blood pressure, the heart rate, and a urine flow rate, and the test data **75** including blood test values such as the renal function, the liver function, and albumin are acquired. In **S105**, a urine flow and creatinine or eGFR are predicted as the effect of the drug. Sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure may be further predicted as the effect of the drug.

[0090] For example, in treatment of heart failure, when a confirmation timing for vital, the urine volume, and the like for the doctor to determine the prescription is delayed due to inconvenience of the doctor's work, a timing for switching the prescription may be delayed, and an organ congestion state may be prolonged. According to the example described above, it is possible to make a relatively quick determination and quickly terminate the organ congestion state. It is not necessary to check the vital, the urine volume, and the like, and it is also possible to reduce the burden on the doctor.

[0091] For example, it is assumed that the renal function of a certain patient is deteriorated and a fluid overload is predominant. For a patient who does not urinate with a normal amount of the diuretic, when the amount of the diuretic is increased, the urine volume can be quickly ensured, but a large increase in the amount of the diuretic can cause dehydration due to rapid water removal, and there is a concern that the renal function is reduced. The urine volume may be ensured by combining with other diuretics, but in this case, an electrolyte abnormality may occur. In the present embodiment, it is possible to predict a transition in the renal function, the electrolyte, the

vital sign, and the like in each prescription candidate, and it can be relatively easy to determine an appropriate amount of the drug and a combination of the drugs.

[0092] As a specific example, in a patient with a reduced renal function, generally, furosemide of a small amount of 5 mg or more and 10 mg or less can be intravenously injected initially, and the amount can be increased while confirming a reaction, and continuous administration or the like is also examined. By applying the present embodiment, it is possible to examine how to administer, by any administration method, an appropriate amount of furosemide from an initial stage of the treatment. By applying the present embodiment, it is possible to quickly release congestion while maintaining a renal function, an electrolyte balance, and the like.

[0093] As described above, in the present embodiment, the control unit **21** of the prescription assistance device **20** receives an input of the time series data **72** on the patient. The control unit **21** receives an input of the attribute data **71** which is basic data on the patient. The control unit **21** receives an input of the drug history data **73** indicating a prescription history of a drug. The control unit **21** generates a plurality of prescription candidates from the time series data **72**, the attribute data **71**, and the drug history data **73**. The control unit **21** infers an outcome of each prescription candidate on the patient from the time series data **72**, the attribute data **71**, and the drug history data **73**. Specifically, the outcome can be an effect of the drug. As an example, the outcome can be a parameter related to the cardiac function and the renal function of the patient and a parameter related to complications associated with the treatment. The control unit **21** outputs an inference result on the outcome.

[0094] In the present embodiment, by selecting a combination of complicated prescriptions from several combinations, it is possible to achieve both simplification of the prescription and improvement of the outcome.

[0095] As a modification of the present embodiment, the control unit **21** may receive, from the user, an operation for adjusting at least one of the plurality of prescription candidates. That is, the user may change a plurality of prescription candidates.

[0096] As a modification of the present embodiment, when a drug is administered to the patient, the control unit **21** may acquire data on the state of the patient at a timing determined for each type of the drug, and may feed the data back to the trained model used in **S105**.

[0097] As a modification of the present embodiment, when a therapeutic intervention different from drug administration is performed on the patient, the control unit **21** may exclude at least a part of the time series data **72** before the therapeutic intervention and predict the effect of the drug. As an example, the therapeutic intervention can include at least one of respiratory support and blood circulation support. The respiratory support can be performed using, for example, a respirator. The blood circulation support can be performed, for example, as a surgery such as coronary artery bypass grafting.

[0098] As a modification of the present embodiment, the control unit **21** may generate a plurality of prescription candidates using a trained model to which the attribute data **71**, the time series data **72**, and the drug history data **73** are input and from which a plurality of prescription candidates are output. The previous prescription may be included in an input of the trained model.

Second Embodiment

[0099] Differences from the first embodiment will be mainly described.

[0100] In the present embodiment, as shown in FIG. 5, the prescription candidate generation function **61** of the prescription assistance device **20** includes, instead of the previous prescription input function **65**, a drug combination setting function **68** and a drug combination storage function **69**.

[0101] An operation of the prescription assistance device **20** according to the present embodiment will be described with reference to FIG. 6. The operation corresponds to a prescription assistance method according to the present embodiment.

[0102] Since the processing of **S201** to **S203** is the same as the processing of **S101** to **S103** in the

first embodiment, the description of the processing of S201 to S203 will be omitted.

[0103] In S204, when the control unit 21 receives an operation for instructing the end of the processing from a user via the input unit 24 such as a touch screen, the control unit 21 ends the processing. When such an operation is not performed, the processing of S205 is performed.

[0104] In S205, the control unit 21 generates a plurality of prescription candidates by the prescription candidate generation function 61.

[0105] Specifically, the control unit 21 sets, as a prescription range by the prescription range setting function 63, a restriction range of a type, an administration amount, and an administration timing of the drug that may be prescribed. The control unit 21 stores the set prescription range in the storage unit 22 by the prescription range storage function 64. The control unit 21 receives, by the drug combination setting function 68, an input of one or more combinations of the type, the administration amount, and the administration timing of the drug from the user via the input unit 24 such as a touch screen. The control unit 21 stores, by the drug combination storage function 69, the input combination in the storage unit 22 as a drug combination designated by the user. The control unit 21 creates, by the prescription candidate creation function 66 based on the drug combination designated by the user that is stored in the storage unit 22, a plurality of prescription candidates by combining the type, the administration amount, and the administration timing of the drug in a plurality of patterns such that the combinations fall within the prescription range stored in the storage unit 22. If the drug combination designated by the user falls within the prescription range, the control unit 21 may adopt the drug combination designated by the user as the prescription candidate as it is. The control unit 21 outputs the plurality of created prescription candidates by the prescription candidate output function 67.

[0106] Since the processing of S206 and S207 is the same as the processing of S105 and S106 in the first embodiment, the description of S206 and S207 will be omitted.

[0107] In S208, the control unit 21 receives, from the user, an operation for selecting one of the plurality of prescription candidates output in S205.

[0108] Specifically, the control unit 21 receives, from the user via the input unit 24 such as a touch screen, an operation for selecting one of the plurality of prescription candidates output to the output unit 25 such as a display.

[0109] In S209 and S210, the control unit 21 controls a device that administers a drug to a patient in accordance with the prescription candidate selected in S208.

[0110] Specifically, the control unit 21 controls the pump 40 such as a smart pump as a device that administers a drug to a patient so that at an administration timing specified by the prescription candidate selected by the user, a drug of a type specified by the prescription candidate is administered to the patient by an administration amount specified by the prescription candidate.

[0111] In S211, the control unit 21 presents a drug administration result in S210 to the user.

[0112] Specifically, the control unit 21 outputs a drug administration result to the output unit 25 such as a display.

[0113] After S211, the processing of S202, S203, and S204 is performed again.

[0114] As described above, in the present embodiment, a user can select a prescription according to an inference result on the outcome. Further, the prescribed drug can be fed to the patient according to a prescription selection result by the user. The prescription may be output to the electronic medical record 77 according to the prescription selection result.

[0115] As a modification of the present embodiment, a timing at which the control unit 21 acquires the time series data 72 in S202 subsequent to S210 may be set according to the prescription applied in S210. The setting may be manually performed by the user or may be automatically performed by the control unit 21. Further, the set timing may be output, as a time, to the output unit 25 such as a display. When the time series data 72 is not acquired at the time or within a certain period of time from the time, a warning may be output to the output unit 25 such as a display.

[0116] As a specific example, it is considered that urine volumes after 15 minutes to 20 minutes

and after 2 hours to 4 hours from a time point at which a loop diuretic such as Lasix® is prescribed are predicted, and actual urine volume data after 15 minutes to 20 minutes and after 2 hours to 4 hours is acquired and fed back.

[0117] As another specific example, it is considered that a urine volume after 5 minutes from a time point at which a vasopressor such as dobutamine is prescribed is predicted, and actual urine volume data after 5 minutes is acquired and fed back to the control unit **21**.

[0118] The present disclosure is not limited to the above embodiments. For example, a plurality of blocks described in the block diagram may be integrated, or one block may be divided. Instead of executing the plurality of processes described in the flowchart in time series according to the description, the processes may be executed in parallel or in a different order as necessary or according to a processing capability of the device that executes each process. In addition, modifications can be made without departing from the gist of the present disclosure.

[0119] The detailed description above describes embodiments of a prescription assistance device and a prescription assistance method. These disclosed embodiments represent examples of the prescription assistance device and the prescription assistance method disclosed here. The invention is not limited, however, to the precise embodiments and variations described. Various changes, modifications and equivalents can be effected by one skilled in the art without departing from the spirit and scope of the invention as defined in the accompanying claims. It is expressly intended that all such changes, modifications and equivalents which fall within the scope of the claims are embraced by the claims.

Claims

1. A prescription assistance device for assisting determination an appropriate prescription of drugs for treatment of heart failure from a variety of combinations of a type, an administration amount, and an administration timing of the drugs comprising: a processor configured to: acquire attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generate, receive, or read, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; predict at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure, based on the attribute data, the time series data, and the drug history data, as an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and present a prediction result of at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure to a user for each of the plurality of prescription candidates via an output interface for determination by the user an appropriate prescription of drugs as in the treatment of heart failure.
2. The prescription assistance device according to claim 1, wherein the attribute data includes at least one of patient's age, sex, blood type, and comorbidity.
3. The prescription assistance device according to claim 1, wherein the drug history data includes at least one of a type, an administration amount, an administration method, an administration start date and time, and an administration end date and time of the drug.
4. The prescription assistance device according to claim 1, wherein the time series data includes test data.
5. The prescription assistance device according to claim 4, wherein the test data includes at least

one of a data of liver function, a data of renal function, creatinine, estimated glomerular filtration rate (eGFR), brain natriuretic peptide (BNP), N-terminal fragment of pro-brain natriuretic peptide (NT-proBNP), hemoglobin, sodium, potassium, blood urea nitrogen (BUN), albumin, and C-reactive protein (CRP).

6. The prescription assistance device according to claim 1, wherein the processor predicts the effect of the drug on renal function, and predicts at least one of urine flow rate, creatinine, and eGFR as the effect on renal function.

7. The prescription assistance device according to claim 1, wherein the drug for each of the plurality of prescription candidates includes at least one of a diuretic, a vasodilator, a cardiotonic, a vasopressor, a heart rate controlling drug, an analgesic, and an infusion.

8. A prescription assistance method for assisting determination an appropriate prescription of drugs for treatment of heart failure from a variety of combinations of a type, an administration amount, and an administration timing of the drugs comprising: acquiring attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generating, receiving, or reading, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; predicting at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure, based on the attribute data, the time series data, and the drug history data, as an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and presenting a prediction result of at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure to a user for each of the plurality of prescription candidates via an output interface for determination by the user an appropriate prescription of drugs as in the treatment of heart failure.

9. The prescription assistance method according to claim 8, wherein the attribute data includes at least one of patient's age, sex, blood type, and comorbidity.

10. The prescription assistance method according to claim 8, wherein the drug history data includes at least one of a type, an administration amount, an administration method, an administration start date and time, and an administration end date and time of the drug.

11. The prescription assistance method according to claim 8, wherein the time series data includes test data, and the test data includes at least one of a data of liver function, a data of renal function, creatinine, estimated glomerular filtration rate (eGFR), brain natriuretic peptide (BNP), N-terminal fragment of pro-brain natriuretic peptide (NT-proBNP), hemoglobin, sodium, potassium, blood urea nitrogen (BUN), albumin, and C-reactive protein (CRP).

12. The prescription assistance method according to claim 8, wherein the processor predicts the effect of the drug on renal function, and predicts at least one of urine flow rate, creatinine, and eGFR as the effect on renal function.

13. The prescription assistance method according to claim 8, wherein the drug for each of the plurality of prescription candidates includes at least one of a diuretic, a vasodilator, a cardiotonic, a vasopressor, a heart rate controlling drug, an analgesic, and an infusion.

14. The prescription assistance method according to claim 8, further comprising: administering at least one of the plurality of prescription candidates to the patient to treat heart failure.

15. A non-transitory computer-readable medium storing a computer program code executed by a computer processor that executes a process for a prescription assistance device for assisting determination an appropriate prescription of drugs for treatment of heart failure from a variety of

combinations of a type, an administration amount, and an administration timing of the drugs, the process comprising: acquiring attribute data indicating an attribute of a heart failure patient, time series data on a state of the patient, including vital measurement values on the patient at a plurality of time points, including at least one of blood pressure, heart rate, urine volume and urine flow rate, and drug history data indicating a prescription history of a drug for the patient; generating, receiving, or reading, a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; predicting at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure, based on the attribute data, the time series data, and the drug history data, as an effect of the drug for each of a plurality of prescription candidates in which combinations of a type, an administration amount, and an administration timing of the drug are different from one another; and presenting a prediction result of at least one of urine flow, creatinine, eGFR, sodium, chlorine, potassium, blood pressure, heart rate, cardiac output, left ventricular end diastolic pressure, left ventricular ejection fraction, and central venous pressure to a user for each of the plurality of prescription candidates via an output interface for determination by the user an appropriate prescription of drugs as in the treatment of heart failure.

16. The computer-readable medium according to claim 15, wherein the attribute data includes at least one of patient's age, sex, blood type, and comorbidity.

17. The computer-readable medium according to claim 15, wherein the drug history data includes at least one of a type, an administration amount, an administration method, an administration start date and time, and an administration end date and time of the drug.

18. The computer-readable medium according to claim 15, wherein the time series data includes test data, and the test data includes at least one of a data of liver function, a data of renal function, creatinine, estimated glomerular filtration rate (eGFR), brain natriuretic peptide (BNP), N-terminal fragment of pro-brain natriuretic peptide (NT-proBNP), hemoglobin, sodium, potassium, blood urea nitrogen (BUN), albumin, and C-reactive protein (CRP).

19. The computer-readable medium according to claim 15, further comprising: predicting the effect of the drug on renal function, and predicting at least one of urine flow rate, creatinine, and eGFR as the effect on renal function.

20. The computer-readable medium according to claim 15, wherein the drug for each of the plurality of prescription candidates includes at least one of a diuretic, a vasodilator, a cardiotonic, a vasopressor, a heart rate controlling drug, an analgesic, and an infusion.
